

JOSHUA NATHANIEL C. TATING

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EDUCATION

De La Salle University

Bachelor of Science in Computer Science, Major in Software Technology

Manila, Philippines

Aug 2022 – Dec 2026

- **Relevant Coursework:** Software Engineering, Web Application Development, Database Systems, Intelligent Systems.
- Consistent Dean's Lister.

EXPERIENCE

Software Engineer | NestJS, Fastify, GraphQL, Next.js, Docker, Prisma, PostgreSQL

Nov 2025 – Present

NGnair Payments

Wyoming, USA

- Own **3** of the platform's core services; engineered a fault-tolerant payment orchestration service integrating **5** payment processors (Elavon, Worldpay, Deluxe) with automatic failover and activation-time routing configuration, eliminating a per-request cross-service round-trip and a class of checkout-blocking failures.
- Delivered an end-to-end integration of a new card processor solo: OAuth2 client-credentials auth with token caching, PCI-scoped hosted payment fields, merchant boarding with identity verification, full payment lifecycle (authorize, capture, refund, void), and fail-closed, idempotent webhook ingestion.
- Redesigned an **AI-assisted merchant underwriting pipeline** into a 3-stage architecture (single-call LLM risk synthesis, deterministic cost gating, code-based policy engine), cutting LLM API costs by **83%** and producing fully replayable audit records for compliance review.
- Built a public API security layer with OAuth2 client-credentials flow, scoped machine JWTs (RS256/JWKS), per-consumer API keys with constant-time validation, and fail-closed token verification enforcing per-client data boundaries.
- Migrated a centralized authentication service from Rails to NestJS, implementing JWT issuance, refresh-token rotation, MFA, SSO token exchange, and standardized RBAC scopes consumed by **9+ microservices** through an Apollo Federation supergraph.
- Extracted checkout and merchant onboarding into independently deployable **Next.js** applications, hardening them with session-bound access control, CSP nonces, and origin-locked iframe **postMessage** embedding for secure cross-origins portal integration.

Freelance Developer | TypeScript, Express.js, Prisma, PostgreSQL, React, MongoDB, Node.js

Mar 2025 – Present

Independent Clients

Manila, Philippines

- Built a transportation booking platform for a logistics company using **Express, TypeScript, Prisma, and PostgreSQL**, with JWT authentication and a documented **26-endpoint OpenAPI** contract.
- Modeled the booking lifecycle as an enforced state machine with terminal-state locking, an edit cutoff window, and audit logging on every status change.
- Engineered a full-stack e-commerce platform using the MERN stack with Material-UI, implementing JWT-based role-based access control for Admin, Staff, and Customer portals.
- Designed a responsive inventory management system with multi-image upload capability and real-time stock tracking, reducing product lookup time by **60%**.
- Built an analytics dashboard with Chart.js visualizations and PDF/CSV export functionality, reducing manual reporting time by **40%**.

PROJECTS

ChronoCritters | Java, Spring Boot, React, TypeScript, gRPC, WebSockets

Sept 2025 – Oct 2025

- Built a distributed microservices system using **Spring Boot** and **gRPC**, reducing code duplication by 40% while maintaining sub-150ms gameplay latency.
- Engineered a WebSocket-based lobby and matchmaking service using **STOMP**, enabling seamless player connectivity and instant battle state synchronization.
- Designed a modular battle engine using the Chain of Responsibility pattern, decoupling complex turn execution logic (Abilities, Effects) and cutting the service by **143 lines**.
- Developed a type-safe React frontend integrated with auto-generated **GraphQL** hooks, ensuring strict contract alignment with the backend schema and preventing runtime errors.

NexusHub | Node.js, Express.js, MongoDB, EJS, JavaScript, CSS

Jan 2024 – Apr 2024

- Engineered a full-stack forum application using **Node.js, Express, and EJS** supporting real-time discussions and role-based authentication (Passport.js) for **100+ registered users**.
- Built admin moderation and data integrity systems, implementing regex-based search filters and MongoDB (Mongoose) cascading deletes to manage **1000+ posts and replies** while reducing moderation time by **40%**.

Socratic AI Tutor (Thesis) | Next.js, PostgreSQL, Vercel AI SDK, LLM Tool Calling

Aug 2025 – Feb 2026

- Built an **agentic LLM tutoring system** using the Vercel AI SDK and Groq API, implementing structured tool-calling to orchestrate reasoning workflows and transition the model from open-ended generation to deterministic misconception detection.
- Designed sequential prompt engineering and evaluation pipelines with in-context reflection constraints, reducing diagnostic hallucinations (False Positive Rate) from **13.3% to 1.3%** and achieving **98.0% precision** in detecting code misconceptions.
- Architected a **Human-in-the-Loop (HITL) LLM validation backend** using Next.js and PostgreSQL to ground responses in verified data and allow educators to govern dynamic context injection.

TECHNICAL SKILLS

Languages & Databases: Java, C/C++, JavaScript, TypeScript, Python, PostgreSQL, MongoDB, Redis, HTML, CSS, Bash

Frameworks & Libraries: Spring Boot, NestJS, Next.js, React, Node.js, Express.js, Fastify, GraphQL, gRPC, Prisma, Lombok, Zustand, Tailwind CSS

Tools & Platforms: Docker, Git, GitHub Actions, Vercel, Render, Vite, Supabase, Jest, Vitest, Playwright, Supertest, Postman, Neovim, VS Code, Arch Linux, Claude Code